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import java.util.List;

// =====
// SUBSYSTEM CLASSES
// FACADE PATTERN
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// These classes handle the complex logic of the photobooth
// - WatermarkEngine: Applies watermarks to images
// - PaymentProcessor: Processes payments
// - CloudStorage: Uploads files to cloud storage
// - PrinterService: Prints images
// =====

class WatermarkEngine {
    public String applyFrame(List<String> images, String templateId) {
        System.out.println("[WatermarkEngine] Applying frame with templateId: " + ...
        System.out.println("[WatermarkEngine] Processing " + images.size() + " ima...
        return "framed_image_" + templateId + ".jpg";
    }
}

class PaymentProcessor {
    public String charge(double amount, String paymentMethod) {
        System.out.println("[PaymentProcessor] Charging " + amount + " via " + pay...
        return "TXN-" + System.currentTimeMillis();
    }
}

class CloudStorage {
    public String upload(String file) {
        System.out.println("[CloudStorage] Uploading file: " + file);
        return "FILE-" + file.hashCode();
    }

    public String generateDownloadLink(String fileId) {
        System.out.println("[CloudStorage] Generating download link for fileId: " ...
        return "https://storage.photobooth.com/download/" + fileId;
    }
}

class PrinterService {
    public void print(String file) {
        System.out.println("[PrinterService] Printing file: " + file);
    }
}

// =====
// FACADE CLASS
// =====

class PhotoboothSessionFacade {
    private WatermarkEngine watermarkEngine;
    private PaymentProcessor paymentProcessor;
    private CloudStorage cloudStorage;

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private PrinterService printerService;

public PhotoboothSessionFacade() {
    this.watermarkEngine = new WatermarkEngine();
    this.paymentProcessor = new PaymentProcessor();
    this.cloudStorage = new CloudStorage();
    this.printerService = new PrinterService();
}

public String processCompleteSession(
    List<String> images,
    String templateId,
    double amount,
    String paymentMethod) {

    System.out.println("\n===== Starting Photobooth Session =====");

    // Step 1: Apply watermark/frame
    String framedImage = watermarkEngine.applyFrame(images, templateId);

    // Step 2: Process payment
    String transactionId = paymentProcessor.charge(amount, paymentMethod);
    System.out.println("[Facade] Payment successful. Transaction ID: " + trans...

    // Step 3: Upload to cloud
    String fileId = cloudStorage.upload(framedImage);
    String downloadLink = cloudStorage.generateDownloadLink(fileId);

    // Step 4: Print the image
    printerService.print(framedImage);

    System.out.println("===== Session Complete =====\n");
    return downloadLink;
}
}

// =====
// CLIENT CLASS
// =====

class Application {
    private PhotoboothSessionFacade facade;

    public Application() {
        this.facade = new PhotoboothSessionFacade();
    }

    public void onCheckoutClicked(List<String> images) {
        // The client only needs to call one method in the Facade
        // without knowing the details of the subsystems inside
        String downloadUrl = facade.processCompleteSession(
            images,
            "TEMPLATE_VINTAGE",
            25000.0,

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        "QRIS");
    System.out.println("[Application] Download your photos here: " + downloadU...
    }
}

// =====
// MAIN - Entry Point
// =====

public class PhotoboothFacadePattern {
    public static void main(String[] args) {
        // Simulating the user pressing the checkout button
        Application app = new Application();

        List<String> capturedImages = List.of(
            "photo_001.jpg",
            "photo_002.jpg",
            "photo_003.jpg");

        app.onCheckoutClicked(capturedImages);
    }
}

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